

How gamma changes access to the target under GD and Adam

Small tanh slopes can leave needed target components difficult to acquire under both gradient descent and Adam. On the same finite dictionary, the explicit spectral comparison below certifies rank-32 eigenvalue-ratio increases of approximately 35, 428, and 36,790 times relative to the actual gamma-8 ratio when gamma rises to 12, 16, and 64. At gamma 8, the chosen slow subspace initially holds 4.33% of the target energy. After 200,000 common-recipe Adam updates, its residual energy alone still exceeds the 1% error tolerance. At larger gammas it is below tolerance.

The distinction between rates and optimizer behavior remains essential. The GD spectrum predicts its training curve exactly, up to numerical evaluation. Adam changes the update geometry, so we measure residual projections and first crossings of smoothed loss. With one shared 100-update EMA half-life, crossings occur at 7,688, 1,900, and 1,486 updates for gammas 12, 16, and 64. Gamma 8 does not cross within 200,000. These are measured Adam times, not predictions from a GD theorem; later recrossings occur. The corrected delay argument below supplies the upper-rate and target-overlap bounds that a necessary GD delay actually requires.

Symbol	Meaning
γ, h	Common tanh slope and center spacing
$J_\gamma, K_\gamma = J_\gamma J_\gamma^T$	Sample-normalized feature matrix and readout kernel
μ_i, u_i	Descending eigenvalues and orthonormal sample-space eigenvectors
$\rho_i = \mu_i / \mu_1$	Learning rate relative to the fastest direction under normalized-step GD
y, r_n	Normalized target samples and residual after n updates
$p_i = u_i^T y ^2 / \ y\ ^2$	Fraction of initial target energy in direction u_i
$P_\gamma(\rho), R_{\gamma,n}(\rho)$	Initial and remaining energy in positive modes with ratios at most ρ
H, T_{EMA}	Adam horizon and first crossing of exponentially averaged squared relative error

1 Why the largest eigenvalue matters

For inputs x_a , centers c_j , and m samples, use ordinary tanh features and a bias:

$$J_\gamma[a, j] = \frac{\tanh(\gamma(x_a - c_j))}{\sqrt{m}}, \quad J_\gamma[a, 0] = \frac{1}{\sqrt{m}}.$$

With normalized targets $y_a = f(x_a) / \sqrt{m}$, minimizing $\frac{1}{2} \|J_\gamma w - y\|^2$ is ordinary half mean-squared error. Geometry is frozen; only readout w is trained. From $w_0 = 0$, GD gives $r_{n+1} = (I - \eta K_\gamma) r_n$. The largest eigenvalue limits the stable step shared by all directions. For $\eta = 1/(2\mu_1)$,

$$\frac{\|r_n\|^2}{\|y\|^2} = \sum_i p_i(\gamma) \left(1 - \frac{\rho_i(\gamma)}{2}\right)^{2n}.$$

A ratio of one halves that component's residual every update. A ratio of 10^{-6} requires about two million updates for one e-fold reduction. A small ratio matters only if the target needs that direction. Both eigenvectors and target weights can change with gamma; ordered rank is not a fixed physical direction across dictionaries. For a prescribed common step, the rate is $\eta\mu_i$, and ratios alone do not determine update counts. The calculations below use the actual archived steps.

2 A finite-dictionary theorem with gamma explicit

This is the cross-gamma comparison from the collaborator's [alternative-view note](#), written with μ for eigenvalues. Centers are consecutive and uniformly spaced; inputs are fixed. Uniform input sampling makes the explicit increment Toeplitz and easy to apply, but the quadratic-form identity itself does not require uniform samples.

Let there be W centers, spacing h , and center-cell interval $I = [c_1 - h/2, c_W + h/2]$. Define $d_{ab} = x_a - x_b$ and

$$K_\gamma^{(0)}[a, b] = \frac{W+1}{m} - \frac{2}{hm} d_{ab} \coth(\gamma d_{ab}), \quad E_\gamma = K_\gamma - K_\gamma^{(0)},$$

using the continuous diagonal value $d \coth(\gamma d) = 1/\gamma$ at $d = 0$. The correction E_γ includes the finite center interval and replacement of a center integral by discrete centers. It is retained. For $\gamma_b > \gamma_a$, set

$$G_{ba}[a, b] = \frac{2}{hm} [d_{ab} \coth(\gamma_a d_{ab}) - d_{ab} \coth(\gamma_b d_{ab})].$$

Theorem 1 (Explicit gamma gain and finite spectral comparison). The matrix G_{ba} is positive semidefinite. For every real sample vector v ,

$$v^T G_{ba} v = \frac{2}{\pi h m} \int_{\mathbb{R}} \frac{M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2}{\omega^2} \left| \sum_a v_a e^{-i\omega x_a} \right|^2 d\omega,$$

$$M_\gamma(\omega) = \frac{z}{\sinh z}, \quad z = \frac{\pi|\omega|}{2\gamma}.$$

Let U_i contain orthonormal eigenvectors for the largest i positive eigenvalues of K_{γ_a} . Set

$$C_i = \text{diag}(\mu_1(\gamma_a), \dots, \mu_i(\gamma_a)) + U_i^T G_{ba} U_i,$$

$$B_i = U_i^T (E_{\gamma_b} - E_{\gamma_a}) U_i, \quad \delta_i = \|C_i^{-1/2} B_i C_i^{-1/2}\|_2.$$

For any $L_b \geq \mu_1(\gamma_b)$,

$$\boxed{\frac{\mu_i(\gamma_b)}{\mu_1(\gamma_b)} \geq \frac{[1 - \delta_i]_+ + \mu_{\min}(C_i)}{L_b}}.$$

Here $\mu_{\min}(C_i)$ is the smallest eigenvalue of the displayed small matrix. The computed bound uses the maximum absolute row sum of the actual finite kernel for L_b .

The multiplier increases with gamma: larger gamma removes less high-frequency content. The theorem carries that gain into an entire trial subspace, then accounts for the finite correction and normalization by the largest eigenvalue. It does not assert that every finite-dictionary ratio increases. An improvement is guaranteed only when the bound exceeds the old ratio. The calculation uses an old eigenspace and small spectral problems; it is not a scalar law depending on gamma alone.

Proof

Write $g_{ab}(c) = \tanh(\gamma(x_a - c)) \tanh(\gamma(x_b - c))$. Direct integration gives

$$\int_{\mathbb{R}} (1 - g_{ab}(c)) dc = 2d_{ab} \coth(\gamma d_{ab}).$$

For distinct inputs, the identity $1 - \tanh u \tanh v = \coth(u-v)(\tanh u - \tanh v)$ reduces the integral to two translated tanh profiles. Their difference integrates to twice the translation. The coincident limit follows from $\int \text{sech}^2(\gamma c) dc = 2/\gamma$.

The center integral over I equals Wh minus the whole-line deficit plus the deficit outside I . Its normalized kernel is therefore $K_\gamma^{(0)}$ plus the omitted-exterior correction. Adding the discrete-center quadrature difference gives exactly $K_\gamma = K_\gamma^{(0)} + E_\gamma$.

The density $s_\gamma(x) = \frac{\gamma}{2} \text{sech}^2(\gamma x)$ satisfies $\tanh(\gamma \cdot) = s_\gamma * \text{sign}$: both derivatives equal $2s_\gamma$ and both vanish at zero. Under convention $\widehat{f}(\omega) = \int f(x)e^{-i\omega x} dx$, substituting $u = e^{2\gamma x}$ and $a = \omega/(2\gamma)$ gives

$$\begin{aligned}\widehat{s}_\gamma(\omega) &= \int_0^\infty \frac{u^{-ia}}{(1+u)^2} du = \Gamma(1-ia)\Gamma(1+ia) \\ &= \frac{\pi a}{\sinh(\pi a)} = M_\gamma(\omega).\end{aligned}$$

This uses the beta integral and gamma reflection identity, with continuous value one at zero. The difference

$$D(d) = 2[d \coth(\gamma_a d) - d \coth(\gamma_b d)]$$

is integrable. The second derivative of $2d \coth(\gamma d)$ is $4(s_\gamma * s_\gamma)(d)$, by differentiating the deficit integral and integrating once by parts. Hence $\widehat{D}'' = 4(M_{\gamma_a}^2 - M_{\gamma_b}^2)$ and

$$\widehat{D}(\omega) = \frac{4(M_{\gamma_b}(\omega)^2 - M_{\gamma_a}(\omega)^2)}{\omega^2}.$$

The continuous zero-frequency limit is $\widehat{D}(0) = \frac{\pi^2}{3}(\gamma_a^{-2} - \gamma_b^{-2})$. Fourier inversion contributes $1/(2\pi)$, giving precisely $2/(\pi h m)$ in the quadratic-form identity.

The larger-gamma multiplier is at least as large, so the transform is nonnegative. Fourier inversion proves the identity and positivity. In particular C_i is positive definite. Compressing the actual new kernel gives $U_i^T K_{\gamma_b} U_i = C_i + B_i$. By definition of δ_i , this is at least $(1 - \delta_i)C_i$ in quadratic-form order. Min-max yields $\mu_i(\gamma_b) \geq (1 - \delta_i)\mu_{\min}(C_i)$ for $\delta_i < 1$. When $\delta_i \geq 1$, positivity of the actual kernel gives zero as a lower bound. Divide by $\mu_1(\gamma_b) \leq L_b$. $\square$

3 From needed directions to a GD delay

Define positive slow energy

$$P_\gamma(\rho) = \sum_{0 < \rho_i(\gamma) \leq \rho} p_i(\gamma).$$

If $P_\gamma(\rho) \geq p > \varepsilon^2$, retaining these terms in the exact GD residual gives

$$E_\gamma(n) \geq \sqrt{p}(1 - \rho/2)^n, \quad n \geq \left\lceil \frac{\log(\sqrt{p}/\varepsilon)}{-\log(1 - \rho/2)} \right\rceil$$

as necessary time to reach relative error ε . For the actual archived step, replace $\rho/2$ by the per-update rate cutoff. The positive restriction separates slow acquisition from permanent nullspace error. The compact-filter calculation in the [structured-kernel note](#) supplies lower bounds on positive slow energy with its declared numerical allowances.

This time bound is a separate target-dependent statement. A lower bound on an eigenvalue ratio establishes a rate floor; it does not by itself establish a necessary delay. The full target-weighted spectrum gives the reference forecast; compact positive-energy bounds give conservative necessary times. The collaborator's larger-gamma comparison is valid; using that lower bound alone to conclude necessary small-gamma delay would reverse the needed inequality.

The missing direction: an upper rate and target-overlap bound

Fix reference slope $\Gamma \geq \gamma$. The same decomposition gives

$$K_\gamma = K_\Gamma - G_{\Gamma, \gamma} + R_{\gamma, \Gamma}, \quad R_{\gamma, \Gamma} = E_\gamma - E_\Gamma.$$

The explicit positive gamma gain from Section 2 is now subtracted. This is the correct direction for proving weak smaller-slope action on needed directions. The correction must remain: $K_\Gamma - G_{\Gamma,\gamma}$ need not itself be positive semidefinite.

Choose any orthonormal trial basis V and set $P = VV^T$. Its columns are candidate slow sample-space directions, not necessarily eigenvectors. Define

$$A_\gamma = V^T K_\gamma V, \quad C_\gamma = (I - P)K_\gamma V, \quad q = \frac{\|V^T y\|}{\|y\|}.$$

A_γ measures action inside the proposed slow subspace; C_γ measures action sent outside it. Both follow from the gamma decomposition including its finite correction. For example,

$$\alpha_\gamma = \mu_{\max}\left(V^T(K_\Gamma - G_{\Gamma,\gamma})V\right) + \|V^T R_{\gamma,\Gamma} V\|_2, \\ \mu_{\max}(A_\gamma) \leq \alpha_\gamma.$$

Retaining the signed correction inside the small matrix can give a tighter upper bound. Likewise retain corrected off-subspace action to obtain $c_\gamma \geq \|C_\gamma\|_2$. Compressed action alone does not establish that trial directions remain slow: coupling is the missing check.

Theorem 2 (Small-gamma slow target energy and necessary GD delay). Let $K_\gamma = J_\gamma J_\gamma^T$, $y \neq 0$, $V^T V = I$, and suppose the bounds above on A_γ and C_γ hold. For $b > \alpha_\gamma$, set

$$\delta = \min\left(1, \frac{c_\gamma}{b - \alpha_\gamma}\right), \quad L = \left[q\sqrt{1 - \delta^2} - \delta\sqrt{1 - q^2}\right]_+^2.$$

At least a fraction L of target energy lies in actual kernel modes with eigenvalues in $[0, b]$. For any explicit readout witness w_* , put

$$e_0 = \frac{\|y - J_\gamma w_*\|^2}{\|y\|^2}, \quad p = [L - e_0]_+.$$

At least p target energy lies in positive modes $0 < \mu_i(\gamma) \leq b$. With zero initialization, $0 < \eta\|K_\gamma\|_2 \leq 1$, and $\eta b < 1$,

$$E_\gamma(n) \geq \sqrt{p} (1 - \eta b)^n.$$

If $p > \varepsilon^2$, reaching relative error ε requires

$$n \geq \left\lceil \frac{\log(\sqrt{p}/\varepsilon)}{-\log(1 - \eta b)} \right\rceil.$$

For a dimensionless ratio upper bound, divide b by a *lower* bound on μ_1 , such as a nonzero Rayleigh quotient. Dividing by Section 2's row-sum upper bound would give the wrong direction. Timing above uses the actual prescribed step and avoids extra normalization.

Proof. Let $F = \mathbf{1}_{(b,\infty)}(K_\gamma)$ project onto true fast modes. With $X = FV$, compressing $K_\gamma V = VA_\gamma + C_\gamma$ gives the Sylvester equation

$$(K_\gamma|_{\text{ran } F})X - XA_\gamma = FC_\gamma.$$

Its solution is the integral of $e^{-tK_\gamma|_{\text{ran } F}} FC_\gamma e^{tA_\gamma}$ over $t \geq 0$. The exponential norms give the integrable factor $e^{-t(b-\alpha_\gamma)}$, hence $\|FV\|_2 \leq \delta$. If $q = 0$ or $\delta = 1$, the bound is zero and immediate. Otherwise normalize y and put $v = Py/q$, $d = \|Fv\| \leq \delta$, and $s = (I - F)v/\sqrt{1 - d^2}$. The unit slow vector s has overlap $\sqrt{1 - d^2}$ with v and overlap at most d with any unit vector orthogonal to v . Decompose $y = qv + \sqrt{1 - q^2}z$ to get

$$\|(I - F)y\| \geq |s^T y| \geq \left[q\sqrt{1 - d^2} - d\sqrt{1 - q^2}\right]_+ \geq \sqrt{L}.$$

The nullspace part of y is orthogonal to every $J_\gamma w_*$, so its squared norm fraction is at most e_0 . Subtracting leaves p in positive slow modes. Each retains residual factor at least $(1 - \eta b)^n$ under the stated step condition. Summing squared residuals proves the error bound; solving for n gives the time bound. $\square$

Eigenvectors are not assumed fixed across gamma. Gamma determines weak action and coupling; the target enters through q and the witness. An easy target can make the conclusion trivial. The target-angle conversion is sharp given its leakage bound, but coupling and kernel bounds can remain conservative. This is a GD theorem, not an Adam rate law.

What the reverse comparison gives numerically

We evaluate actual finite tanh action on trial spaces and audit the explicit gamma gain and finite correction. We neither discard the correction nor substitute a periodic spectrum. Candidate spaces are tails of a fixed gamma-64 reference basis and eigenbands of the kernel projected onto discrete polynomial spaces of degrees 32, 64, and 128. The latter require a small spectral solve at each gamma. Polynomial truncation selects a trial space; omitted action remains in C_γ .

A fixed spectral-cutoff grid selects the strongest necessary bound without reading observed GD acquisition times. Taking the maximum valid lower bound over these spaces is valid. The default adds explicit floating-point allowances to compressed upper rates and coupling, with zero-, one-, and ten-times allowance sensitivity. These are FP64 evaluations of an exact-arithmetic theorem, not interval-certified numerical bounds.

Strongest necessary times at gammas 8, 12, 16, and 64 are **6,458,359; 169,485; 34,555; and 5,927 updates**, compared with actual crossings of **15,798,313; 186,057; 61,792; and 16,013**. They retain about **41%, 91%, 56%, and 37%** of measured delay. Degree-128 projected spaces win at the first three gammas; a reference tail wins at gamma 64. Earlier compact-filter bounds remain separate.

Fixed reference directions are substantially less informative at small gamma: they give only 361, 644, and 951 updates at gammas 8, 12, and 16. The useful bound retains changing finite geometry and target alignment. This repairs the logical direction and improves measured delay bounds, but does not create a scalar formula in gamma alone.

4 What we measure for Adam

[Adam](#) rescales readout-coordinate updates and uses gradient history. For $r_n = Jw_n - y$, its update is

$$r_{n+1} = r_n - \alpha_n J D_n \widehat{m}_n, \quad D_n = \text{diag}((\sqrt{\widehat{v}_n} + \epsilon_{\text{Adam}})^{-1}).$$

Raw-kernel eigenmodes are diagnostic coordinates, not independent Adam decay modes. Even without momentum, the effective kernel would be the time-dependent $J D_n J^T$. We do not apply the GD time law to Adam.

Instead, at saved checkpoints measure

$$R_{\gamma,n}(\rho) = \sum_{0 < \rho_i(\gamma) \leq \rho} \frac{|u_i(\gamma)^T r_n|^2}{\|y\|^2}.$$

Initially this equals $P_\gamma(\rho)$. At a fixed update count, it asks whether Adam removed target components occupying the GD-slow subspace. A substantial remaining value supports the proposed obstruction for this optimizer and protocol. A small value indicates acquisition of those components; that outcome must also be reported.

First EMA acquisition (main Adam metric). Let $\ell_n = E(n)^2$ be squared relative error. With half-life $s = 100$ updates and $\beta = 2^{-1/s}$, define

$$M_0 = \ell_0 = 1, \quad M_n = \beta M_{n-1} + (1 - \beta) \ell_n, \\ T_{\text{EMA}} = \min\{n \leq H : M_n \leq 10^{-4}\}.$$

Plots show $\sqrt{M_n}$ in relative-error units. Averaging squared error matches training loss; this is not an EMA of the error norm. Initialization retains the true initial loss instead of a falsely low average. Crossing means first acquisition of the smoothed-loss threshold, not permanent settlement. Runs without a crossing are censored at $H = 200,000$.

The half-life is one shared, post-hoc plotting choice, not a gamma-specific tuning. Sensitivity uses 10, 30, 100, 300, 1,000, and 3,000 updates. Metrics use every recorded update, not sparse-checkpoint interpolation. No optimizer reruns or settings changes are involved.

The EMA itself has a resolution limit. Nonnegative losses and $M_0 = 1$ imply $M_n \geq \beta^n$. Even zero loss after initialization cannot cross before $\lceil s \log_2(10^4) \rceil$ updates: **1,329 updates at half-life 100**. The gamma-64 crossing at 1,486 is close to this smoothing floor. EMA compresses differences between sufficiently fast runs; it is not an exact optimizer time. The sensitivity figure explicitly displays this floor.

Sustained acquisition (supporting diagnostic). With the same horizon and $\varepsilon = 0.01$,

$$T_{\text{sust}} = \min\{n : E(k) \leq \varepsilon \text{ for all } k = n, \dots, H\}.$$

If $E(H) > \varepsilon$, acquisition is unobserved within the horizon. For a successful run, this is one plus the last update above threshold, or zero when no update is above threshold. Report the confirmation length $H - T_{\text{sust}} + 1$. No smoothing is used and nothing is asserted beyond H .

5 Matched experimental protocol

The primary target is $f(x) = \sin(2\pi x) + \frac{1}{2} \sin(6\pi x) + \frac{1}{4} \sin(10\pi x)$ on $[-1, 1]$. The archive uses spacing $h = 1/256$, 559 tanh features plus bias, and 8,193 uniformly spaced training inputs. Centers, samples, target, raw readout coordinates, and zero initialization are fixed; slopes are 8, 12, 16, and 64.

Common-recipe Adam uses initial rate 10^{-3} , additive epsilon 10^{-12} , and moments (0.9, 0.999). Its rate is constant through 20,000 updates, decays by cosine to 10^{-3} of its initial value at 50,000, then remains fixed. Runs continue to 200,000 updates. This schedule is a material part of the acquisition result.

Sensitivity comparisons reuse validation-selected recipes: five initial rates crossed with two epsilons, selected by median validation error at 40,000, 42,500, 45,000, 47,500, and 50,000 updates; optimizer states continue intact. No new selection uses present spectral projections. All plots concern training-grid optimization; separate validation data selected the sensitivity recipes. No new held-out generalization claim is made.

Supporting targets are $\exp(\sin(3\pi x))$, $1/(1+25x^2)$, $\sqrt{5}x^2$, and $\sqrt{2}\sin(2\pi x)$. Targets and checkpoints are not selected for agreement with the hypothesis. These scalings are retained because Adam's additive epsilon makes scaling part of the protocol. The saved SVD cutoff is numerical, not an exact-rank statement. Omitted energy is reported as unresolved, not automatically as nullspace.

6 What the matched measurements show

The main comparison is shown on the following landscape page so that the spectral and acquisition-time panels remain legible.

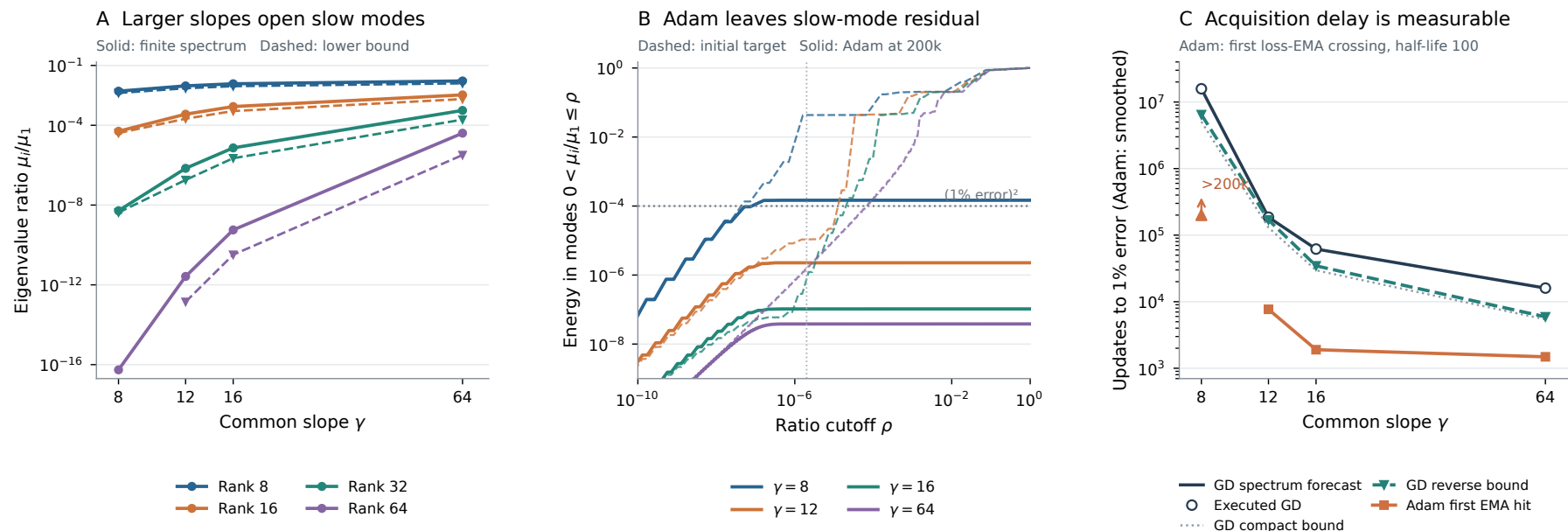


Figure 1: Gamma, needed directions, and acquisition. A: finite-kernel ratios and numerical larger-gamma rate lower bounds, referenced to gamma 8. B: initial target energy and common-recipe Adam residual energy after 200,000 updates, accumulated over resolved positive modes. Energy 10^{-4} corresponds to 1% relative error. C: executed GD crossings, full-spectrum forecasts, necessary GD times from the reverse-subspace and earlier compact bounds, and first Adam loss-EMA crossings with shared 100-update half-life. GD has longer executed trajectories; the Adam arrow denotes its own budget. Target and geometry match across panels. Panel A's rate lower bounds do not supply panel C's necessary times; those require positive slow-energy lower bounds. No theoretical time curve is asserted for Adam.

The spectral theorem is informative, with measurable slack

At rank 32 the old ratio is 5.2174×10^{-9} . Bounds at gammas 12, 16, and 64 are 1.8172×10^{-7} , 2.2340×10^{-6} , and 1.9195×10^{-4} , retaining about 26%, 30%, and 35% of actual new ratios. The bound is within a factor of four at this prescribed rank while exposing a large explicit gamma effect. Rank 64 is less tight: bounds retain only 5.3%, 5.7%, and 8.0%. The gamma-8 baseline rank-64 bound is numerically unresolved and labeled accordingly. Ranks are not selected after seeing tightness.

The slow subspace explains the remaining gamma-8 error

At illustrative cutoff $\rho = 2 \times 10^{-6}$, gamma 8 has initial positive slow energy 0.0433333. After 200,000 common-recipe Adam updates, the remaining energy is 1.4709443×10^{-4} , alone exceeding the 10^{-4} budget for 1% error. Total relative error is 0.0121283; 99.9998% of residual energy lies in that band. Adam removed about 99.66% of its initial energy, but the remainder prevents acquisition.

At the same cutoff, final slow residual energies are 2.2660×10^{-6} , 1.0437×10^{-7} , and 3.8307×10^{-8} at gammas 12, 16, and 64, below tolerance. The cutoff is a descriptive probe inherited from the earlier $\eta\mu \leq 10^{-6}$ analysis under $\eta\mu_1 \approx 1/2$. Main cumulative curves expose full cutoff dependence. The subspaces share a spectral definition across gamma, not the same eigenvectors.

Initial slow energy is not monotone: it is 2.85×10^{-7} at gamma 16 and 1.48×10^{-6} at gamma 64. Adam can increase a diagnostic subspace’s residual energy: at 20,000 updates the gamma-12 and gamma-16 slow energies exceed their initial values. These are projected residuals, not necessarily untouched initial target content.

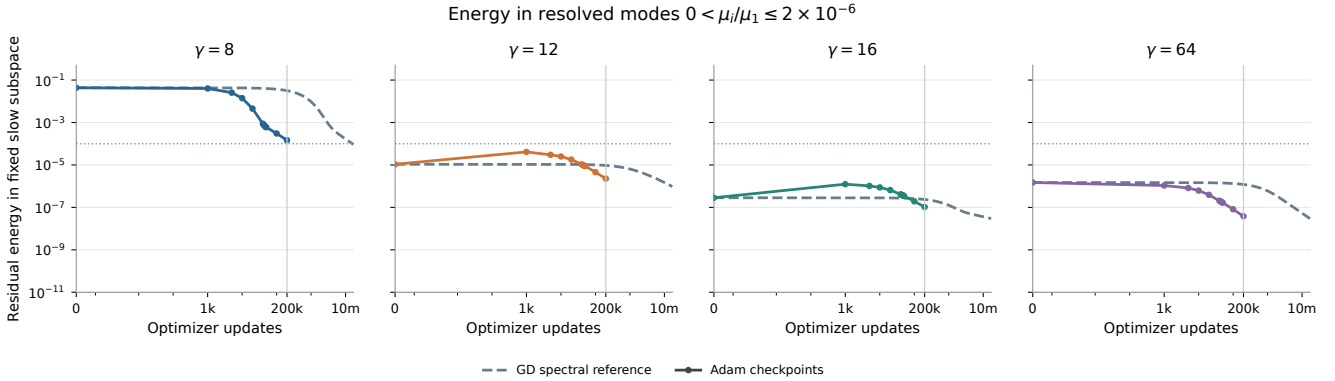


Figure 2: Evolution within the slow band. The cutoff is 2×10^{-6} for every gamma. GD curves follow its target-weighted spectrum; Adam values are saved readout checkpoints projected onto each finite kernel’s modes. Energies use the original target norm. Lines between Adam checkpoints do not determine exact acquisition time; every-update traces do.

First EMA crossings reveal early acquisition

Common Adam first crosses raw 1% at 4,084, 1,361, and 333 updates for gammas 12, 16, and 64. Loss-EMA first crossings are 7,688, 1,900, and 1,486. This resolves earlier learning concealed by the previous sustained-crossing plot. EMA later recrosses above threshold 73, 97, and 98 times: first acquisition is not settled acquisition. Raw sustained crossings remain 41,397, 41,412, and 41,539, reflecting oscillations and rate decay. Gamma 8 never crosses within 200,000 under either recipe.

Window sensitivity matters. Half-life 30 gives common crossings 4,374, 1,430, and 510; half-life 300 gives 26,465, 11,278, and 5,342. At half-life 1,000 they cluster near 28,000–30,000 and lose monotone ordering. Long memory

and oscillations can restore a nearly flat comparison. Selected-recipe half-life-100 crossings are 40,208, 41,716, and 10,076. The evidence supports the stated target, recipe, and metric, not universal monotonic Adam improvement.

Executed GD reaches 1% at 15,798,313, 186,057, 61,792, and 16,013 updates. Compact necessary times are 5,072,048, 130,057, 29,640, and 5,421. The full constructed-spectrum forecast agrees with executed crossings; conservative bounds retain about 32%, 70%, 48%, and 34% of observed delay. These times do not follow by substituting panel A's ratio lower bounds.

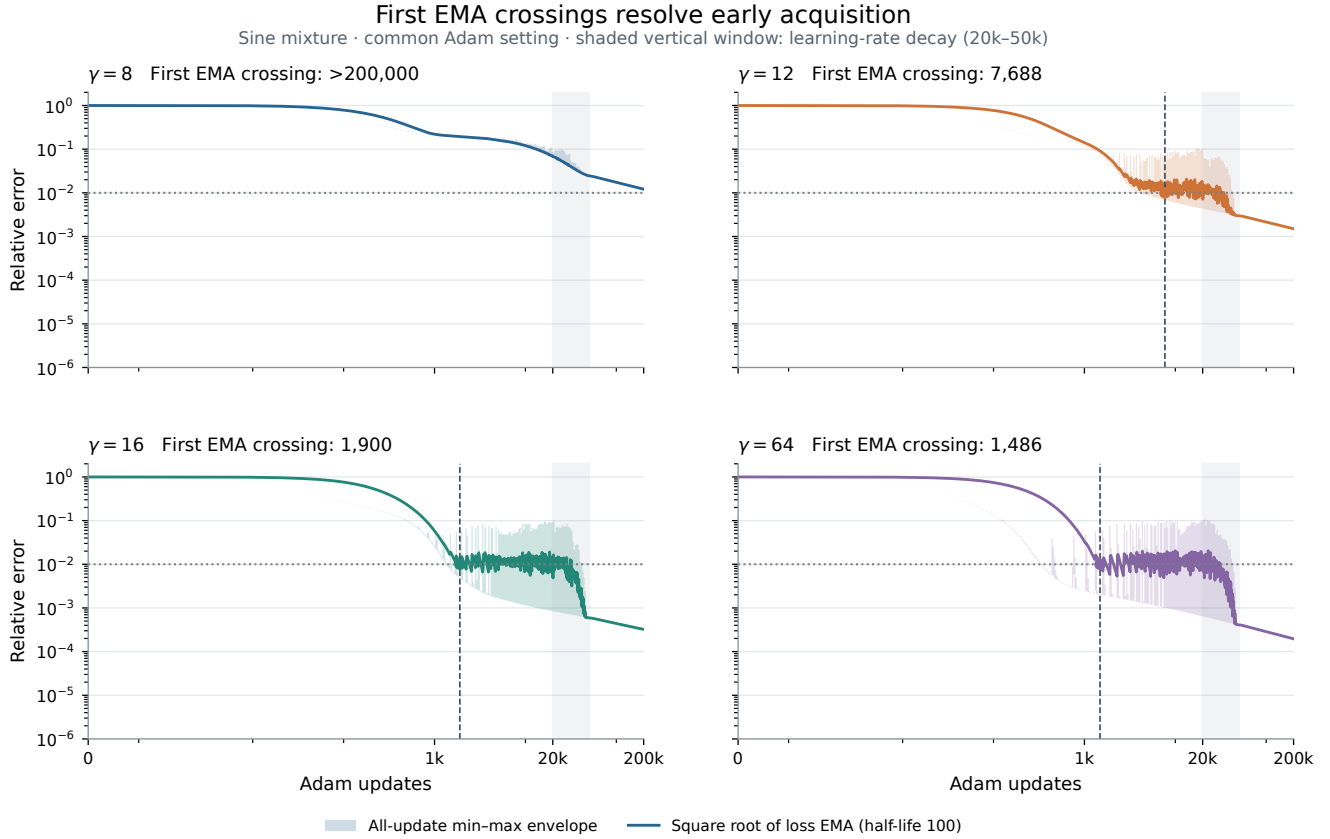


Figure 3: Where the Adam crossing comes from. Colored bands contain minima and maxima of actual per-update errors within update bins. Solid curves show the square root of loss EMA with shared 100-update half-life. The dashed vertical line and threshold dot mark its first crossing, computed from every update. Gray windows mark rate decay from 20,000 to 50,000. All curves come from saved training data.

Five-target sensitivity prevents a universal claim. In selected Runge runs at gammas 12 and 16, the first 1% crossing is update 122; last recrossings postpone sustained acquisition to 199,808 and 199,710. Only 193 and 291 confirming states remain through the endpoint. Common-recipe single-sine final error also does not improve monotonically with gamma. These exceptions are retained.

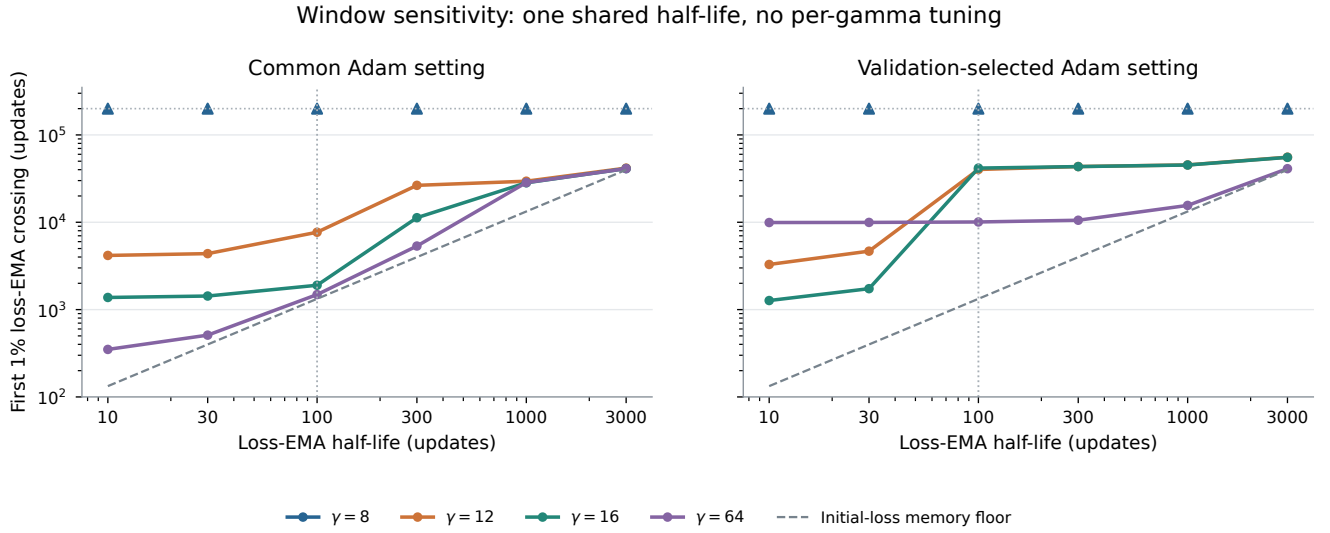


Figure 4: The smoothing choice is visible. Every gamma uses the same half-life at each horizontal position. The dotted line marks the main choice 100; triangles at 200,000 updates denote no observed crossing. The gray dashed curve is the earliest crossing permitted by memory of the initial loss, even if all later losses were zero. Long windows materially change acquisition times. These are sensitivity calculations on the same trajectories, not new training.

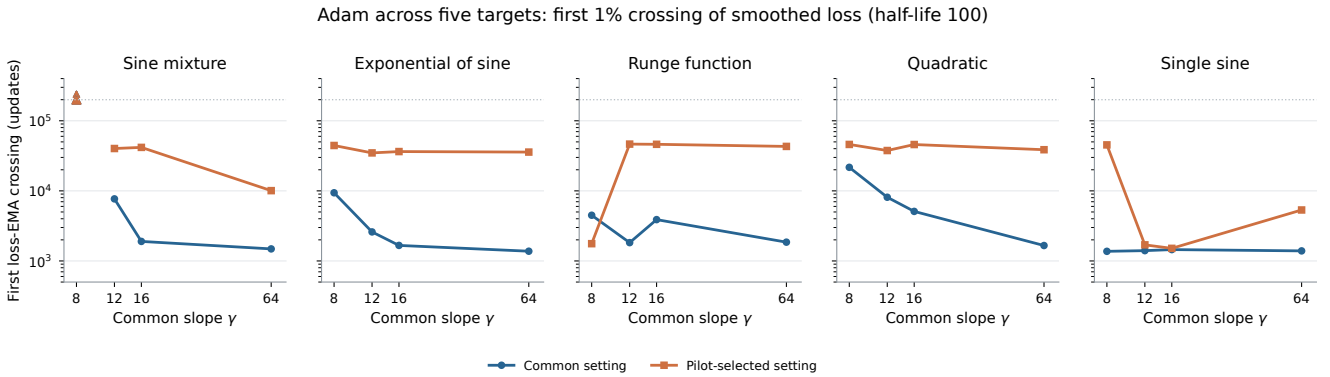


Figure 5: Target and recipe sensitivity. Every archived target is included with the same first loss-EMA crossing and 100-update half-life. Recipes were selected by original late-window validation error, not fastest acquisition or agreement with this kernel argument. Arrows denote no EMA acquisition within 200,000 updates.

7 Numerical verification and reproduction

Adam analysis covers 40 cases and 12 saved checkpoints per case. Direct feature reconstruction matches archived hashes. All 480 residual norms agree with saved measurements within 9.14×10^{-16} . The most negative total-minus-projected energy is -2.89×10^{-15} , consistent with floating-point subtraction; this signed discrepancy is retained. Omitted-SVD energy is monitored, not labeled irreducible error.

A fresh rectangular SVD and direct sample-space residual projection reproduce band energies across all 480 checkpoint/target/view cases at seven ratio cutoffs. The largest band-energy discrepancy is 5.02×10^{-14} and energy closure differs by at most 5.33×10^{-14} . At the displayed cutoff, common-recipe discrepancies are below 3.54×10^{-16} . At gamma 16 the recomputed numerical retained rank is larger by three, and at gamma 64 by one. These are tiny-cutoff numerical rank changes, not a change in the true feature space, and do not affect the 2×10^{-6} band. The gamma-8 obstruction persists at the stricter cutoff 2×10^{-7} , where remaining energy is 1.45561×10^{-4} .

A fresh read-only persistent-archive audit processed 400 trace chunks and all states 0 through 200,000 for the 40 selected series. First, last-above, and sustained crossings agree exactly with both remote and local arrays at all six recorded tolerances. Remote metadata and local compact-manifest hashes agree. This used one CPU for 30 seconds and no GPU. Only requested scalar errors were retrieved; old data was unchanged.

The explicit Toeplitz increment agrees with independently computed direct rows within 5×10^{-16} . The small-matrix calculation checks positivity resolution, old trial-space gaps, relative correction conditioning, and two whitening formulations. These are floating-point checks, not outward-rounded certificates for new ratio bounds. The implemented C_i uses $U_i^T K_{\gamma_a} U_i$ instead of the equivalent eigenvalue diagonal to preserve the measured trial-space identity.

All 23 focused tests pass, including rotated-trial-space reverse bounds and an independent explicit EMA recurrence. The reverse-action reconstruction discrepancy is below 8.5×10^{-20} in Frobenius norm; this checks algebraic consistency, not the finite correction independently. Under zero-, one-, and ten-times arithmetic allowances, gamma-8 degree-128 necessary times are 6,554,361; 6,458,359; and 5,539,258. Gamma 12 changes only from 169,486 to 169,482 over this range. Allowance sensitivity does not constitute a rounding certificate.

EMA analysis recomputes all 40 complete trajectories at six half-lives. Its compact artifacts and the reverse-bound evidence reproduce byte-for-byte. An independent figure audit matches every target-case crossing, every window-sensitivity crossing, and every plotted EMA curve to those artifacts. Curves retain each exact first crossing and its preceding state; sparse visual sampling can omit later oscillations, whose counts come from complete traces.

The full non-slow suite reports 833 passed, 17 failed, 9 skipped, and 4 deselected. The 17 failure identifiers exactly match the pre-existing baseline; no new test failure was introduced.

The [gamma_optimizer_access_evidence](#) contains source hashes, array conventions, unresolved statuses, retained ranks, and every plotted quantity. Full scalar traces remain in the persistent archive; compact envelopes and the independent audit accompany this study. Reproduce locally with one BLAS thread:

```
export OPENBLAS_NUM_THREADS=1 VECLIB_MAXIMUM_THREADS=1
python -m experiments.expD36_frozen_gamma_probe.gamma_ratio_bound
python -m experiments.expD36_frozen_gamma_probe.gamma_reverse_bound
python -m experiments.expD36_frozen_gamma_probe.adam_spectral_analysis
python -m experiments.expD36_frozen_gamma_probe.adam_projection_audit
python -m experiments.expD36_frozen_gamma_probe.adam_ema_analysis \
    --archive /path/to/adam_raw_scalar_archive.npz
python -m experiments.expD36_frozen_gamma_probe.optimizer_access_figure
latexmk -pdf -outdir=/tmp/gamma-optimizer-latex \
    docs/gamma_optimizer_access_note.tex
```

The optional archive audit is in `adam_trace_audit.py`: `extract` reads the persistent run directory; `verify` compares the scalar archive with local compact evidence. No least-squares refit, spectral solve, or detached diagnostic changes a trained readout. No new training was performed.